

EPISTAXIS IN CHILDREN

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INTRODUCTION

1. Epistaxis occurs commonly in children, especially in those between the ages of 2 and 10 years.
2. In most cases, nosebleeds are secondary to local trauma and can be cared for by primary care physicians.
3. In rare instances, however, a nosebleed may be difficult to control or a manifestation of a serious systemic illness.

EPIDEMIOLOGY

1. Estimated 30% of children have one nosebleed by the time they are 5 years of age.
2. In children between the ages of 6 and 10 years, the frequency increases to 56%.
3. Nosebleeds are rare in infancy and infrequent after puberty.
4. An increased incidence of epistaxis occurs during hot or cold weather and when ambient humidity is low, all of which make the nasal septal mucosa dry and friable, and predisposed to bleeding, even with minor trauma as may occur with nose rubbing, blowing, or sneezing.

HISTORY

1. Age
 - Rare in children younger than 2 years (approximately 1 per 10,000), should prompt consideration of trauma (intentional or unintentional) or serious illness (e.g. thrombocytopenia)
 - Recurrent idiopathic epistaxis appears to resolve with time and is uncommon in children older than 14 years
2. Duration of bleeding
 - Prolonged bleeding may suggest bleeding disorder
3. Bilateral / unilateral
 - Unilateral bleeding: may be an isolated lesion / minor trauma
 - Bilateral: more suggestive of general mucosal irritation, systemic aetiology, or major nasal trauma
4. What measures were taken to stop the bleeding?
 - Bleeding that is difficult to control with anterior pressure may indicate a bleeding disorder or posterior source of bleeding
5. History of trauma, including nose picking
 - Nasal trauma usually causes epistaxis that in most cases resolves spontaneously
6. History of nasal congestion, discharge, or obstruction
7. History of foreign body insertion
 - E.g. button battery
 - May present with persistent unilateral bloody / foul discharge
8. Ongoing nasal discharge
 - May suggest allergic rhinitis
9. Any regular medication (such as anticoagulants, long term aspirin)

PHYSICAL FINDINGS

1. Look for possible hemodynamic compromise, signs of systemic causes of bleeding, and asphyxiation
 - Malaise (may suggest hypoxia $\pm$ acidosis after asphyxiation)
 - Pallor (may indicate blood loss or anaemia)
 - Petechiae, bruising, gingival bleeding, hemotympanum (think of trauma)
 - Mucocutaneous telangiectasias, haemangiomas
 - Enlarged lymph nodes, organomegaly, petechiae, pallor (may indicate haematological disease / malignancy)
 - Icterus $\pm$ hepatomegaly (sign of liver disease with secondary coagulopathy)
 - To do visual acuity and extraocular movements (if history of facial trauma)
2. Nasal examination (most bleeds happen anteriorly)
 - Any active bleeding, scabs, ulcerations, erosions, or prominent blood vessels over the Little's area
 - Foreign body
 - Masses (tumours)
 - Rhinitis (infectious / allergy)
 - Signs of allergy (pale / bluish nasal mucosa and oedematous, boggy nasal turbinates)
 - Vascular anomalies (telangiectasia, haemangioma) - presence of telangiectasia in nose / oral cavity and lips suggest hereditary haemorrhagic telangiectasia

RED FLAGS

- ▶ If epistaxis is uncontrolled (does not stop after 20 minutes) with simple first aid measures
- ▶ Presence of systemic features such as fever, other mucocutaneous bleeding

INVESTIGATIONS

1. Laboratory evaluation is not indicated in majority of children with self-limited epistaxis.
2. If systemic causes are suspected, consider doing full blood count (FBC), peripheral blood film (PBF), and coagulation screen.
3. Nasal bone X-ray is indicated for suspected nasal bone fracture, for medicolegal issues such as assault, road traffic accidents, suspected NAI.

DIFFERENTIAL DIAGNOSES

Local Causes

1. Trauma
 - From nose picking
 - Inflammation of nasal mucosa from upper respiratory infection
 - Foreign bodies
 - External trauma from fall or blunt force
 - Non-accidental trauma (consider especially if child younger than 2 years)
2. Allergic Rhinitis (AR)
 - AR with inflammation & subsequent drying of the mucosa may lead to epistaxis
 - Airborne environmental pollutants have been associated with increased inflammation of the nasal mucosa
 - Use of intranasal corticosteroids (ICS)
 - may cause epistaxis due to steroid's direct drying effect on the nasal septum or local trauma from administration of the medication

- however, in a small, randomized trial in 2023, of 60 children with untreated AR & first presentation of epistaxis, those receiving ICS with or without antihistamines had significantly lower severity and frequency of subsequent epistaxis and higher rates of resolution
 - the dispenser tip of nasal sprays may traumatise the already dry and friable mucosa
3. Neoplasms
- Nasal masses
 - Haemangioma in nose
 - Juvenile nasopharyngeal angiofibroma
 - Rare: rhabdomyosarcomas, lymphomas, squamous cell carcinomas of the nose, sinuses or nasopharynx

Systemic Causes

1. Bleeding disorders
 - Thrombocytopenia (idiopathic thrombocytopenic purpura, leukaemia, aplastic anaemia, HIV infection)
 - Inherited (von Willebrand disease, Haemophilia, Glanzmann thrombasthenia, Bernard-Soulier syndrome)
 - Acquired coagulopathy (hepatic disease, severe vitamin K deficiency, malabsorption syndrome)
2. Hereditary blood vessel disorder
 - Hereditary haemorrhagic telangiectasia (Osler-Weber-Rendu disease)
3. Medications
 - Accidental ingestion of aspirin, NSAIDs, warfarin
4. Hypertension
 - Rare in children
5. Inflammatory diseases
 - Wegener granulomatosis, lethal midline granuloma (rare idiopathic inflammatory diseases that lead to nasal tissue destruction and bleeding)

MANAGEMENT

First Aid for Epistaxis (Advice to Parent)

1. Lean forward and spit out any blood
 - Sit the child upright in a comfortable position and lean forward
 - Avoid lying down or tilting the head backwards, as this will lead to swallowing of blood. Swallowed blood irritates the stomach and may cause the child to vomit it out
 - Prepare a basin for the child to spit out any blood that has dripped from the nose into his or her mouth
2. Pinch the soft part of the nose
 - Tightly pinch the soft (not bony) part of the nose, just above the nostrils, for 10 minutes
 - The nostrils should be closed firmly
 - Do not release the pressure to check for bleeding until the 10 minutes is up
 - During this time, child will have to breathe through his or her mouth
 - As an additional step (and not a substitute for manual compression), place an ice pack over the forehead or suck ice cubes
3. If bleeding does not stop after 20 minutes of direct pressure, proceed to Children's Emergency
 - Continue applying pressure to the soft part of the nose in the meantime

Things to Take Note

1. Pressing on the bony part of the nose does not stop a nose bleed
2. Avoid packing the child's nose with anything as bleeding usually recurs when the packing is removed
3. Once the bleeding has stopped, avoid picking at the nose or blowing out the blood clots as this may cause the bleeding to happen again
4. If needed, tell the child to blow his or her nose gently

Prevention of Epistaxis

1. Applying a small amount of petroleum jelly twice a day to the centre wall (septum) inside the nose
 - Helpful for relieving dryness and irritation
2. Use of air humidifier
 - It may be beneficial to use a humidifier in the bedroom at night to moisten the air
3. Put two to three drops of warm water into each nostril before blowing a stuffy nose
 - May also use a saltwater nasal spray
4. Take antihistamines if the child has nasal allergies
 - May help to break the itching-bleeding cycle
 - Consider intranasal steroids spray if allergic rhinitis is suspected
 - Advise not to rub or blow nose
5. Avoid aspirin
 - Aspirin can increase the tendency of the body to bleed easily for up to a week and can make nose bleeds last much longer
6. Consider short term topical decongestants e.g. Oxymetazoline drops for 5 days
 - Causes vasoconstriction to alleviate or prevent recurrence in the acute phase

WHEN TO REFER

Most episodes of epistaxis resolve with compression of nasal alae for 5 to 10 minutes and do not require specialty care.

Referral to emergency department or ENT specialist is indicated for patients with:

1. Refractory epistaxis
 - E.g. uncontrollable bleeding, posterior epistaxis, or hemodynamically unstable
2. Local abnormalities
 - E.g. tumours, telangiectasias
3. Recurrent epistaxis with no apparent cause

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